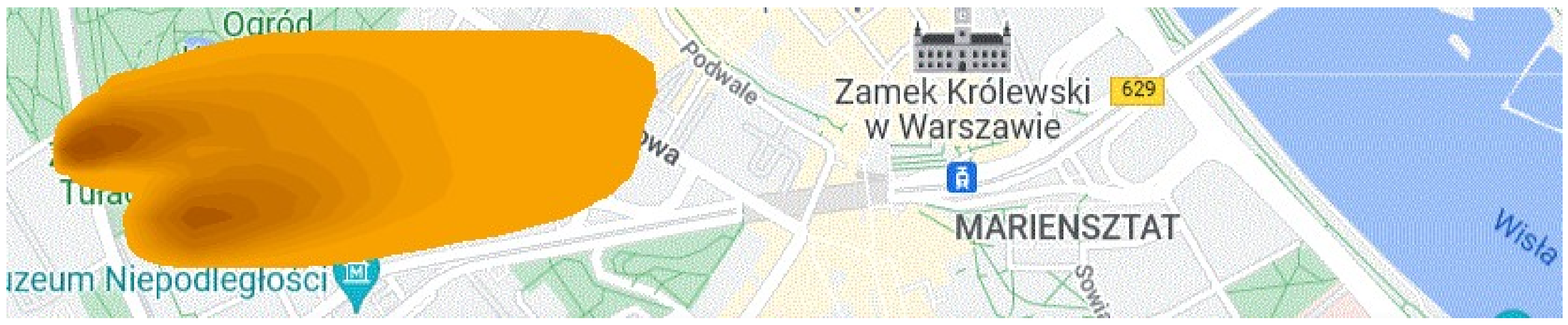
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Project page: https://marek-ac.meri.pl/

Additional results, solver documentation, and interactive demo are available on the project page.



Vanadis: A Directional Residual Stabilization Framework for Large-Scale Environmental Transport Simulations

Three-dimensional air dispersion modelling using the Finite Element Method

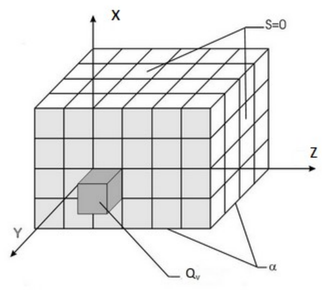
Aim

The goal of this work is to develop a **stable and efficient solver** for large-scale advection–diffusion problems relevant to environmental transport. The proposed **Directional Residual Stabilization (DR)** improves numerical stability in advection-dominated regimes **without introducing artificial damping**, which is crucial for accurate pollutant transport modelling in complex atmospheric flows.
The governing advection–diffusion–reaction equation is formulated in the standard finite-element weak form and integrated in time using the **BDF2 scheme**. The DR term is applied only along the local flow direction, enhancing stability while preserving the physical structure of the equation and enabling efficient CPU/GPU implementation.

Three-dimensional advection–diffusion equation and discretized domain representation.

$$\frac{\partial S}{\partial t}+v_x\frac{\partial S}{\partial x}+v_y\frac{\partial S}{\partial y}+v_z\frac{\partial S}{\partial z}=\frac{\partial}{\partial x}\left(K_x\frac{\partial S}{\partial x}\right)+\frac{\partial}{\partial y}\left(K_y\frac{\partial S}{\partial y}\right)+\frac{\partial}{\partial z}\left(K_z\frac{\partial S}{\partial z}\right)-Q_v+PS$$

The governing equation describes the time evolution of a scalar quantity S (e.g., pollutant concentration) under advection, diffusion, and source terms. Velocity components (v_x,v_y,v_z) drive directional transport, while diffusion coefficients (K_x,K_y,K_z) control spatial smoothing. The diagram illustrates the finite-element discretization of the domain, where each cell represents a control volume with local fluxes Q_v, a reaction/decay term P, Dirichlet boundaries S=0 on outer walls, and a Robin flux αS on the ground surface (deposition). This structure forms the basis for the element-by-element (EBE) implementation used in Vanadis.

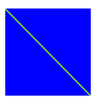


Method

The Vanadis solver uses a finite-element weak formulation with **BDF2 time integration**. The DR stabilization isolates only the residual component aligned with the flow direction, suppressing oscillations while preserving cross-wind gradients. This flow-aligned correction improves stability while maintaining physical fidelity.

Implementation

The solver employs an **element-by-element (EBE)** strategy that avoids global matrix assembly, reduces memory usage, and improves data locality. GPU acceleration is achieved through **CUDA kernels** operating on element-level data, with CPU and GPU pathways sharing identical numerical logic. The resulting system matrix is **diagonally dominant**, making it well suited for **GPU-based DBCG**.
A lightweight diagonal preconditioner is used to maintain robustness while keeping GPU kernels simple and efficient.



Diagonal structure of the system matrix

Mass Preservation

The DR formulation preserves the total transported mass throughout the simulation. Because stabilization is applied **only in the flow direction**, no artificial diffusion is introduced, and plume evolution remains physically consistent. Mass preservation is maintained across both CPU and GPU implementations.

Table 1. Total mass evolution with DR stabilization — source term: 1 kg/s

Time (s)	Mass (kg)	Interpretation
100	100.17	Linear growth
200	201.62	Linear growth
400	413.06	Transient
800	721.12	Approaching equilibrium
1200	796.59	Steady-state region
1600	801.29	Plateau (<0.3% variation)
2000	797.95	Stable equilibrium

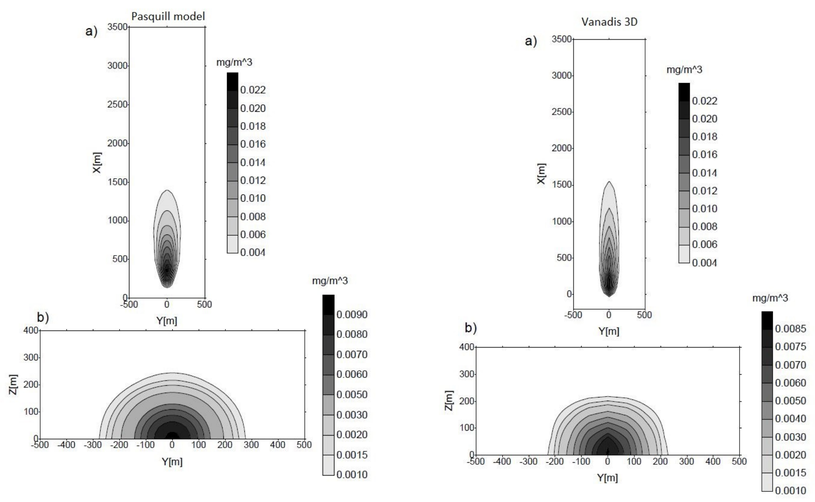
Innovations

- Peclet-dependent directional weighting functions
- Three independent nonlinear stabilizers
- Element-embedded stabilization preserving classical FEM matrix structure
- Scalability to multi-million-element domains
- Hardware-independent CPU/GPU solver behaviour

Results

Physical Validation — Pasquill Comparison

Comparison with the **Pasquill dispersion model** shows consistent plume behaviour and concentration ranges, confirming the physical correctness of the DR-stabilized formulation. The solver remains stable in strongly advection-dominated regimes.



Performance Test — Large Mesh (2.8M Nodes)

A separate performance benchmark was conducted on a 2.8-million-node mesh with 20 time steps (0–2000 s, Δt = 100 s). Vanadis completes the simulation in: • 10 minutes on a budget-class GTX 1060 GPU • 20 minutes on an i5-4590 CPU
During the transient, the number of DBCG iterations decreases from **76** → **36**, with per-step solve times improving from: • **8.9 s** → **6.6 s** on GPU • **20.8 s** → **9.6 s** on CPU
Transient iterations refer to per-step DBCG iterations during the time-dependent phase, while the 16 iterations in Table 2 correspond to the final steady-state solve.
At steady state, the CPU and GPU runs converge to **the same solution**, with identical iteration counts, residual norms, and total mass. This confirms that the DR formulation is fully hardware-independent and that the GPU implementation preserves the algebraic structure and convergence behavior of the CPU solver.

Table 2. CPU vs GPU consistency for DR stabilization

Property	DR – CPU	DR – GPU
Iterations (steady state)	16	16
Residual norm	2.85×10−9	2.85×10−9
Steady-state mass	246.44594	246.44594
Mass conservation	Excellent	Excellent
Numerical behavior	Stable, robust	Stable, robust

These results demonstrate that the solver is efficient even on modest hardware.

Two Chimneys Example

The results are visualized on two orthogonal surfaces: the ground surface and a vertical cross-section. The example highlights the solver's ability to handle **time-dependent source terms**, **changing wind fields**, and **spatially varying parameters**. The simulation also demonstrates the capability to model **complex terrain features**, including the influence of **tall buildings** on flow patterns and pollutant dispersion.



Conclusions

The Vanadis solver provides a **robust, mass-preserving, and physically consistent** framework for large-scale advection–diffusion simulations. Directional residual stabilization ensures stability without artificial damping, while GPU acceleration enables efficient computation across architectures. The method is suitable for **environmental modelling, pollutant transport**, and other **advection-dominated applications**.